

BUSS386 Problem Set 5 — Solutions

Swaps

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Problem 1 — Cheapest-to-deliver & futures invoice

- (a) Cost of delivery = quoted price – settlement $\times$ CF:

Bond	Quoted	92.00 $\times$ CF	Cost
A	99.00	96.60	2.40
B	142.00	140.76	1.24 CTD
C	119.00	116.84	2.16

Bond B is the CTD.

- (b) Invoice cash = settlement $\times$ CF + accrued interest = $92.00 \times 1.5300 + 2.50 = 140.76 + 2.50 =$
 $\boxed{\$143.26}$ per \$100 face.
- (c) The CF is defined as the price of the deliverable bond *at the benchmark yield* (6% for US T-bond futures). Without a fixed benchmark, CFs would be ambiguous and the invoice scaling would not put bonds with different coupons on a comparable footing.

Problem 2 — SOFR futures hedge

- (a) **Long** the futures. SOFR futures price = 100 – rate, so *long* gains when rates *fall*. The depositor loses when rates fall (lower interest), so a long futures position offsets that risk.
- (b) \$100M/\$1M per contract = $\boxed{100 \text{ contracts long}}$.
- (c)
- Futures move: $96.80 - 96.50 = +0.30$ (30 bp). Long P&L = $100 \times 30 \times \$25 = \boxed{+\$75,000}$.
 - Interest on deposit: $\$100\text{M} \times 3.20\% \times 0.25 = \boxed{\$800,000}$.
 - Total proceeds: $\$800,000 + \$75,000 = \boxed{\$875,000}$.
- (d) Implied locked-in rate: $\frac{\$875,000}{\$100,000,000 \times 0.25} = 0.0350 = \boxed{3.50\%}$, exactly the day-1 forecast.

✓

Problem 3 — IRS pricing from the yield curve

- (a) Discount factors today:

$$e^{-0.025} = 0.9753, e^{-0.056} = 0.9456, e^{-0.090} = 0.9139, e^{-0.124} = 0.8834.$$

$$\sum = 3.7182. \quad c = \frac{1 - 0.8834}{3.7182} = \frac{0.1166}{3.7182} = \boxed{3.14\%}.$$

- (b) One year later, with rates at $\{3.0\%, 3.3\%, 3.5\%\}$:

$$e^{-0.030} = 0.9704, e^{-0.066} = 0.9361, e^{-0.105} = 0.9003. \quad \sum = 2.8068.$$

$$V = 100M \cdot [0.0314 \times 2.8068 + 0.9003 - 1] = 100M \cdot (-0.0116) = \boxed{-1.16M}.$$

- (c) Rates have risen, so the new fair fixed coupon (around 3.4%–3.5%) is higher than the locked $c = 3.14\%$. The receiver is stuck collecting a below-market coupon for the remaining 3 years $\Rightarrow$ negative value.

Problem 4 — Currency swap (USD/KRW)

- (a) $F_t = 1,380 \cdot e^{-0.015t}$:

t (yr)	0.5	1.0	1.5	2.0
F_t (₩/USD)	1,369.7	1,359.5	1,349.3	1,339.2

- (b) Method 1 (weighted average of forwards). KRW discount factors at $r = 2.5\%$:

$$e^{-0.0125} = 0.9876, e^{-0.0250} = 0.9753, e^{-0.0375} = 0.9632, e^{-0.0500} = 0.9512; \quad \sum = 3.8773.$$

$$K = \frac{0.9876(1,369.7) + 0.9753(1,359.5) + 0.9632(1,349.3) + 0.9512(1,339.2)}{3.8773} = \boxed{1,354.5 \text{ per USD}}.$$

Method 2 (equivalent): $K = 1,380 \cdot (e^{-0.020} + e^{-0.040} + e^{-0.060} + e^{-0.080}) / 3.8773 = 1,380 \cdot 3.8059 / 3.8773 = 1,354.5. \checkmark$

- (c) $K < S_0 = 1,380$. USD pays a higher interest rate than KRW ($r_f > r$), so the forward USD price (in KRW) must *decline* over time to prevent arbitrage between holding USD vs. KRW.
- (d) The strip of 4 forwards gives *different* KRW payments each period (1,369.7, 1,359.5, 1,349.3, 1,339.2). The swap converts these into a *constant* ₩1,354.5 every period — same PV, smoother cash flow.

Problem 5 — Comparative advantage and the IRS

- (a) **Absolute advantage:** AAA in both markets (lower in fixed by 1.50%; lower in floating by 0.70%).

Comparative advantage: AAA in *fixed* (the gap is wider: 1.50% vs. 0.70%). BBB in *floating* (relatively cheaper there).

(b) Total savings = $1.50\% - 0.70\% = \boxed{0.80\%}$ per year on \$100M (\$800,000 / yr).

(c) Dealer takes 0.10%; remaining 0.70% split equally $\rightarrow 0.35\%$ each.

- **AAA** target net cost = $(\text{SOFR} + 0.10\%) - 0.35\% = \text{SOFR} - 0.25\%$.

Cash flows: pays 4% to its bondholders, receives Y from dealer, pays SOFR to dealer.

$$\text{Net} = 4\% - Y + \text{SOFR} = \text{SOFR} - 0.25\% \Rightarrow \boxed{Y = 4.25\%}.$$

- **BBB** target net cost = $5.50\% - 0.35\% = 5.15\%$.

Cash flows: pays $\text{SOFR} + 0.80\%$ to its bondholders, pays X fixed to dealer, receives SOFR from dealer.

$$\text{Net} = (\text{SOFR} + 0.80\%) - \text{SOFR} + X = 0.80\% + X = 5.15\% \Rightarrow \boxed{X = 4.35\%}.$$

- **Dealer** keeps $X - Y = 4.35\% - 4.25\% = 0.10\%$. ✓

All-in costs: AAA pays SOFR -0.25% (saved 0.35% vs. direct floating); BBB pays 5.15% (saved 0.35% vs. direct fixed); dealer earns 0.10%.

(d) If BBB's credit deteriorates, the next quarterly reset on its floating loan will charge a wider spread, eroding (or eliminating) BBB's savings on the trade.

Problem 6 — Duration-based hedging with KTB futures

(a) Portfolio dollar duration today: $D_P V_P = 6.0 \times 100\text{B} = 600\text{B}$ per 100% yield move. Target: $3.0 \times 100\text{B} = 300\text{B}$. The futures position must therefore contribute $\boxed{-300\text{B}}$ — negative dollar duration, so the fund goes **short**.

(b) Value per contract: $V_F = 1.1050 \times 50\text{M} = 55.25\text{M}$, so each contract carries

$$D_F \cdot V_F = 8.0 \times 55.25\text{M} = 442\text{M}.$$

$$N^* = \frac{300\text{B}}{442\text{M}} = 678.7 \quad \Rightarrow \quad \boxed{\text{short 679 contracts}}.$$

(c) 679 contracts carry $679 \times 442\text{M} = 300.12\text{B}$ against the $\text{W}300\text{B}$ target — an over-hedge of $\boxed{0.04\%}$. Rounding down to 678 would give $\text{W}299.68\text{B}$, an under-hedge of 0.11%, so 679 is the nearer choice. Either way a small residual is unavoidable: the contract size quantizes the hedge.

(d) Selling KTBs would realize accounting gains or losses, incur bid-ask spreads on a large block, and disturb the portfolio's chosen credit and liquidity profile. A futures overlay adjusts duration *synthetically* in one trade, leaves the cash bonds untouched, and unwinds cheaply when the manager wants the original duration back.